CSC483/583 Programming Project: Building (a part of) Watson with a Three-Stage Retrieval Pipeline for Jeopardy!-Style Wikipedia Answer Selection

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Abstract

We study a retrieval formulation of Jeopardy!-style question answering in which each clue must be mapped to the title of a Wikipedia page. The project uses approximately 280,000 Wikipedia pages, and evaluation is performed on 100 Jeopardy clues whose gold answers are Wikipedia titles. We present a three-stage retrieval pipeline that uses increasingly expensive models only after the candidate set has been reduced. Stage 1 uses a weighted ranking of sparse and dense retrieval signals to produce an initial candidate list. Stage 2 reranks the top 100 candidates with a cross-encoder using natural-question reformulations of the clue. Stage 3 reranks only the top 10 candidates from the previous stage with Qwen3-Reranker-4B, a larger reranker model. This staged design improves the P@1 score from 0.31 after the initial weighted retrieval stage to 0.41 after cross-encoder reranking and to 0.47 after the final Qwen reranker. The main takeaway is that strong answer selection comes from combining broad early retrieval with progressively stronger semantic rerankers over much smaller candidate sets.

1 Introduction

Jeopardy!-style clue answering is a difficult retrieval problem because the clue is often indirect, category-dependent, and phrased differently from the text that appears in the target article. Rather than asking for a direct fact, many clues refer to an entity through paraphrase, role, quoted language, or historical description. This creates a lexical mismatch problem: the words in the clue are often not the words that appear most prominently in the correct Wikipedia page.

In this project, we treat the task as page retrieval over Wikipedia. Given a Jeopardy category and clue, the goal is to predict the Wikipedia page title that matches the gold answer. This avoids full generative question answering and turns the task into candidate generation and reranking over approximately 280,000 pages. The main challenge is to retrieve the right page despite noisy Wikipedia markup, long article bodies, redirect pages, and clues whose phrasing differs sharply from natural encyclopedia prose.

Our core design principle is to use simpler and cheaper methods early, then apply larger semantic models later only to narrowed candidate sets. The resulting system is a three-stage retrieval pipeline. Stage 1 uses weighted hybrid retrieval, which combines BM25 ranking over cleaned article text with Dense Passage Retrieval (DPR) embeddings [Karpukhin et al., 2020] indexed in FAISS [Johnson et al., 2019]. Stage 2 reranks the top 100 candidates using a cross-encoder trained for passage ranking [Nogueira and Cho, 2019]. Stage 3 exports only the top 10 candidates from the combined second-stage ranking and reranks them with Qwen3-Reranker-4B, a larger reranker model [Qwen Team, 2025]. We also reformulate each clue into five natural-language questions so that the dense retriever and rerankers see inputs that better match question-versus-passage training distributions. Throughout the paper, Precision@1 (P@1) is the main evaluation metric because the system must return one final Wikipedia title as its answer.

This staged architecture yields consistent gains across the pipeline. The weighted hybrid retrieval stage reaches a P@1 score of 0.31, the combined second-stage ranking improves this to 0.41, and the final Qwen reranker reaches 0.47. These results support a simple conclusion: retrieval quality matters first, but stronger semantic reranking becomes effective once the correct page has been brought into a small candidate set.

2 Task and Dataset

The task in this project is to answer Jeopardy-style clues by retrieving the Wikipedia page whose title matches the gold answer. Each example contains a category, a clue, and one correct Wikipedia title. This makes the problem a page-ranking task rather than a fully generative question answering task: given the category and clue, the system must rank candidate Wikipedia pages and place the correct title at the top.

The dataset provided in the project consists of 100 Jeopardy clues whose answers appear as Wikipedia page titles, together with a Wikipedia collection of approximately 280,000 pages. Each Wikipedia page must be treated as a separate document in the index. This setup creates two main challenges. First, the collection is large enough that retrieval quality matters immediately. Second, Jeopardy clues are often indirect and category-dependent, so the wording of the clue may differ substantially from the wording used in the correct Wikipedia article.

3 Core Implementation

3.1 Wikipedia Preprocessing and Indexing

The Wikipedia collection contains approximately 280,000 pages distributed across 80 raw files. Each page begins with a title in double-square-bracket format, which allows us to parse the corpus into page-level documents rather than indexing each raw file as a single unit. We store parsed pages in SQLite, keep redirect pages separate from article pages, and clean article bodies by removing wiki markup, templates, section noise, and other non-content tokens.

This preprocessing supports the structure of the final task: the system must return a single Wikipedia page title as the answer. Redirects are therefore handled explicitly rather than treated as normal answer pages, and all retrieval stages operate over canonical article identities.

The final first-stage retriever uses one main sparse index and several dense indexes. The sparse component is a Whoosh BM25F index over cleaned full article bodies. The dense component is built from DPR passage representations, with vectors stored in FAISS for efficient nearest-neighbor search. We retain additional lead-only indexes and redirect/title signals in the codebase for experimentation, but they are not part of the final selected weighted configuration.

3.2 Stage 1: Weighted Hybrid Retrieval

The first stage of the final system is a weighted hybrid retriever. It combines one strong sparse signal with three complementary dense signals:

- full cleaned body BM25
- DPR retrieval for the category+clue query
- DPR retrieval averaged over five generated natural questions
- DPR retrieval for the concatenation of those five natural questions

For the selected run, the sparse query uses the category concatenated with the clue text. The active weights are heavily dominated by the full-body BM25 score, with smaller contributions from the dense components. Concretely, the selected configuration uses body weight 15.00 and weight 1.00 for each active dense component.

This stage is designed for broad recall and cheap filtering. It searches the whole collection, produces the first global ranking, and provides the candidate pool for the more expensive rerankers. The result is a hybrid retriever that is substantially stronger than plain sparse retrieval alone.

3.3 Why We Convert Clues into Natural Questions

Jeopardy clues are often not written like ordinary search questions. They may be fragments, indirect descriptions, or clues whose meaning depends heavily on the category. DPR, cross-encoders, and reranker-style models are generally trained on more standard query-document or question-passage pairs. To reduce this mismatch, we rewrite each clue into five natural-language questions using Qwen3-4B [Qwen Team, 2025].

The natural-question prompt is constrained to preserve the same intended answer without introducing outside knowledge. These question reformulations are then used in two places: as dense DPR queries in the first stage and as query inputs for the cross-encoder and Qwen reranker in later stages. This lets the later models compare more natural question forms against Wikipedia evidence, which is particularly useful when the category contributes the answer type or hidden constraint.

3.4 Stage 2: Cross-Encoder Reranking

The second stage reranks the top 100 candidates from the weighted hybrid retriever using a cross-encoder model, `cross-encoder/ms-marco-MiniLM-L-12-v2`. Each candidate article is represented by its title and cleaned article body, split into 100-token chunks. For each of the five generated natural questions, the system scores all chunks and keeps the best chunk score for that question. The final cross-encoder score for the candidate is the average across the five natural questions.

We then combine the cross-encoder score with the initial first-stage score. Both scores are normalized within the top-100 candidate set, and the selected combination uses a weight of 1.00 for the initial score and 7.00 for the cross-encoder score. This stage is still local and relatively efficient because it only reads 100 candidates per clue, but it is much more semantically expressive than stage 1.

3.5 Stage 3: Qwen3-Reranker-4B

The final stage is an offline reranking experiment performed on the top 10 candidates from the combined second-stage ranking. For each clue, we export the top 10 candidates together with the category, clue, gold answer, five generated natural questions, and a truncated evidence snippet from each candidate article. These exported rows are then scored in Colab with Qwen3-Reranker-4B.

The Qwen reranker sees an instruction, one natural question, and one document evidence snippet. It outputs a yes/no-style relevance score, and the final candidate score is the average over the five natural questions for that clue. This stage cannot recover a correct answer that was not already present in the exported top 10, so it is a pure answer-selection stage rather than a recall-improvement stage.

4 Experimental Setup

We evaluate on 100 Jeopardy clues whose gold answers correspond to Wikipedia page titles. The document collection is the full Wikipedia corpus used in the project, containing approximately 280,000 pages after

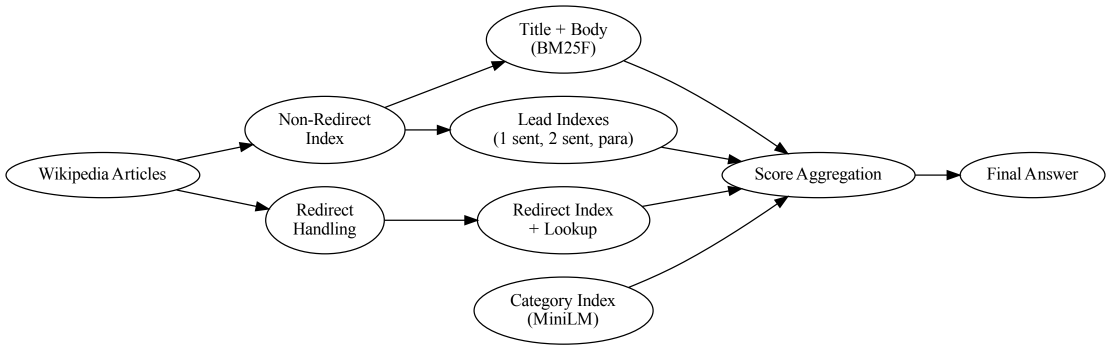

Figure 1: Final three-stage pipeline. A Jeopardy category and clue are first mapped to a weighted hybrid candidate ranking over Wikipedia. The top 100 candidates are reranked by a cross-encoder, the top 10 candidates from the combined second-stage ranking are reranked by Qwen3-Reranker-4B, and the highest-ranked final page title is returned as the answer.

parsing. Each page is indexed as a separate document.

The final selected local pipeline uses the following configuration:

- category + clue query text
- weighted hybrid retrieval as the first stage
- cross-encoder reranking over the top 100 candidates
- natural-question averaging for the cross-encoder query mode
- Qwen3-Reranker-4B reranking over the top 10 candidates from the combined second-stage ranking

We evaluate by normalized title match between the top-ranked predicted page and the gold answer. The primary metric is Precision@1 (Top-1 accuracy), because the system must commit to one final answer. We report Precision@5 as a supporting metric to show whether the correct page is at least being brought near the top of the ranking before the final answer decision.

5 Results and Analysis

5.1 Main Results

Table 1 presents the main system-stage comparison. Precision@1 is the primary metric because it directly measures whether the final predicted page title is correct. Precision@5 is included only as supporting context.

The largest early gains come from better retrieval. Plain Whoosh body-only retrieval performs poorly, reaching only a P@1 score of 0.06. A stronger sparse baseline raises this sharply, and the full weighted hybrid configuration reaches 0.31. This shows that most of the early improvement comes from stronger candidate generation rather than from later reranking alone.

The cross-encoder then improves answer selection inside the retrieved set. Using only the cross-encoder ranking over the top 100 candidates reaches a P@1 score of 0.40, and the combined second-stage ranking

Table 1: Main system-stage comparison. Precision@1 is the primary metric; Precision@5 is supporting.

System stage	P@1	P@5
Plain Whoosh body-only	0.06	0.14
Sparse body-only baseline, no category	0.21	0.42
Sparse body-only baseline + category	0.22	0.46
Weighted hybrid retrieval + category	0.31	0.50
Cross-encoder only	0.40	0.62
Combined first-stage + cross-encoder	0.41	0.61
Qwen chat verifier	0.04	0.29
Qwen3-Reranker-4B	**0.47**	**0.58**

Table 2: Top-1 correct counts after grouping the 30 Jeopardy categories into 9 super-categories.

Super-category	N	Vanilla	Weighted	Cross-enc.	Fused	Qwen
People / Biography	19	2	9	10	11	14
Geography / Places	17	0	4	4	5	8
History / Politics	25	0	4	10	8	9
Entertainment / Pop Culture	14	1	6	7	7	5
Business / Companies	5	0	2	0	0	2
Media / Journalism	5	1	2	2	2	3
Organizations / Institutions	3	1	2	2	2	2
Language / Wordplay	3	0	0	0	0	1
Mixed / General Knowledge	9	1	2	5	6	3
Total	**100**	**6**	**31**	**40**	**41**	**47**

reaches 0.41. This indicates that many first-stage errors are ranking mistakes rather than complete retrieval misses.

The final Qwen reranker gives the best Top-1 result, improving the P@1 score from 0.41 to 0.47. However, this gain should be interpreted correctly: Qwen3-Reranker-4B only sees the exported top 10 candidates from the combined second-stage ranking. It improves final answer selection inside that top-10 set, but it does not improve recall beyond what earlier stages already retrieved.

One negative result is also important. The Qwen chat verifier performs very poorly, reaching only a P@1 score of 0.04. A yes/no chat probability is therefore not a useful final ranking signal for this task, even when the underlying model is much larger than the cross-encoder.

5.2 Grouped Category Analysis

To make the category-level behavior interpretable, we group the 30 original Jeopardy categories into 9 super-categories. Table 2 shows Top-1 counts at each stage.

The grouped analysis shows that the weighted hybrid retriever is especially important for *People / Biography*, *Entertainment / Pop Culture*, and *Geography / Places*. These categories often depend on identifying a named entity, role, or location from article-body evidence, which is well matched by a strong sparse retriever with dense support.

The cross-encoder helps most on *History / Politics*, where the jump from 4 to 10 correct answers is the largest stage-to-stage improvement in the grouped table. These clues often require better local semantic alignment between a clue-like question and a supporting passage. The cross-encoder is much less reliable on *Business / Companies*, where semantically related company, product, and subsidiary pages are strong distractors.

Qwen reranking helps most on *People / Biography* and *Geography / Places*, where it adds three more correct top-ranked answers in each group. This suggests that once the correct answer is already inside the top 10 and the evidence snippet is explicit, the larger reranker can make better final distinctions than the cross-encoder alone. In contrast, *Entertainment / Pop Culture* and *Mixed / General Knowledge* become worse under Qwen reranking, indicating that the final yes/no-style evidence judgment is less stable for broad, indirect, or comparative clues.

5.3 What Did Not Work

Several explored ideas did not become part of the final system. Naively adding category text to plain Whoosh hurt badly, reducing the P@1 score to 0.02, which shows that raw category terms can distort simple sparse search when not balanced properly. Lead-only indexes based on the first sentence, first two sentences, or first paragraph all underperformed full-body retrieval as standalone signals. Title-only and redirect-only retrieval also failed as standalone rankers.

We also explored clue paraphrasing for sparse retrieval, but it did not help enough. The paraphrases tended to preserve the same key nouns and proper nouns while only changing surface wording, so they did not create enough new lexical evidence for BM25-style matching. Finally, several other implemented components, including qwen-summary dense indexes and category-semantic variants, remained at zero weight in the selected configuration because they did not improve the final weighted hybrid pipeline enough to justify inclusion.

6 Conclusion

This project shows that Jeopardy-style Wikipedia answer selection benefits from a staged retrieval design in which stronger models are applied only after the candidate set has already been narrowed. The first-stage weighted hybrid retriever is responsible for the largest early gains because it brings the correct page into a competitive rank range. The cross-encoder then improves semantic ranking within the top 100 candidates, and the final Qwen3-Reranker-4B stage further improves answer selection within the top 10 candidates from the combined second-stage ranking.

The final system reaches a P@1 score of 0.47, compared with 0.31 after initial weighted hybrid retrieval and 0.41 after the combined cross-encoder stage. This supports a clear design lesson: retrieval quality comes first, but larger semantic rerankers become valuable once the system has already produced a strong candidate set at low cost.

The remaining failures follow a small number of patterns. Some questions are still true retrieval misses, where the correct page is not present in the candidate set. Others are ranking misses caused by broad-topic distractors or semantically related wrong pages. Wordplay categories remain difficult because the clue may have little direct lexical overlap with the answer page. Finally, the Qwen reranker is limited by the exported top-10 evidence set, so it cannot recover answers that earlier stages failed to surface.

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